

Capstone Project Proposal

Predictive Maintenance for KPC Pipeline Pump Infrastructure

1. Introduction

Team NULL_TERMINATORS is a five-member team from the KPC Cohort, Inuka Fellowship, Power Learn Project. The team placed first in Hackathon 1, developing a revenue reconciliation pipeline that addressed billing and payment discrepancies in KPC's order-to-cash process. This capstone extends the same reconciliation methodology to a second operational problem within the pipeline network: unplanned pump failure.

Name	Role
Silas Kibet	Data Engineering Lead — pipeline, synthetic data generation, ETL
Brian Kioko	Modelling and Machine Learning Lead — hypothesis testing, feature engineering, modelling
Ingrid Miriam	Dashboard and Visualisation Lead — dashboarding, stakeholder views
Eugene Obunde	Storytelling and ROI Lead — briefs, executive narrative, business case
Lameck Mugo	Quality Assurance and Documentation Lead — testing, UAT notes, deployment readiness

2. The Problem

KPC operates approximately 1,342 kilometres of pipeline connecting Mombasa, Nairobi, Nakuru, Eldoret, and Kisumu, with a combined storage and handling capacity of approximately 14 billion litres annually. Unplanned pump failures at booster and depot stations result in throughput stoppages, delayed deliveries to Oil Marketing Companies, emergency repair costs, and safety-critical pressure excursions.

Documented incidents illustrate the operational risk this presents. An estimated 400,000 litres of product were lost at Thange River in 2015, and 551,000 litres were lost at Kiboko in 2018, valued at approximately KES 63 million. These incidents indicate that undetected mechanical degradation carries measurable safety and financial consequences.

Current maintenance practice is time-based and follows a fixed interval. This approach does not account for the actual condition of individual pumps, resulting in unnecessary servicing of functioning equipment and insufficient attention to units approaching failure.

3. The Solution

The team proposes a condition-based maintenance system that replaces fixed-interval servicing with continuous, data-driven risk assessment. The system operates as follows:

- A classification model evaluates each pump daily against a seven-day failure risk window, using rolling-window sensor features derived from vibration, temperature, pressure, and motor current readings.

- A regression model estimates Remaining Useful Life, in hours, for pumps identified as high-risk, providing maintenance planners with a defined service window.
- Maintenance scheduling is prioritised according to model output, directing technician time and spare-parts allocation toward pumps that require intervention.

This methodology has been validated in the Week 6 laboratory exercise, achieving 94.2 percent classification accuracy, 84.3 percent failure-detection sensitivity, a 2.4 percent false alarm rate, and a Remaining Useful Life prediction error of 105.9 hours (Mean Absolute Error), using a synthetic dataset of 24 pumps across four depots.

4. The Business Value

The value drivers below are structured around the operational levers this system directly affects. Quantified savings will be finalised in collaboration with KPC once historical maintenance and failure-cost data are available; the performance figures cited are those already validated in the Week 6 laboratory exercise.

- Reduced emergency repair expenditure: shifting failures from unplanned to planned reduces reliance on emergency parts sourcing, technician overtime, and expedited logistics.
- Reduced throughput-stoppage duration: earlier detection narrows the window between failure onset and intervention, limiting delivery delays to Oil Marketing Companies.
- Efficient technician deployment: a 2.4 percent false alarm rate limits unnecessary dispatch to functioning equipment.
- Earlier intervention window: daily risk assessment provides a multi-day advantage over a periodic manual inspection cycle.
- Reduced safety exposure: fewer undetected pressure and mechanical anomalies of the type associated with the incidents referenced in Section 2.

5. Go-to-Market Strategy

Adoption of this system is structured in three phases:

- Phase 1, Pilot deployment: a single booster or depot station with sufficient sensor history is selected to validate model performance under real operating conditions.
- Phase 2, Fleet-wide deployment: the system is extended across all ten pumping stations, with station-specific recalibration where operating conditions differ materially.
- Phase 3, Methodology extension: the underlying approach is extended to other infrastructure-dependent utilities facing comparable maintenance challenges, building on the reconciliation methodology previously validated in Hackathon 1.

6. Implementation Rollout

This section describes how the team would deploy the solution to a client, rather than how the team builds the solution internally. The methodology is structured in five phases and is designed to be repeatable across any prospective client site:

Phase	Activity
1. Discovery and Data Assessment	Audit the client's existing pump fleet, sensor infrastructure, and historical maintenance records to determine data availability and identify gaps that must be addressed before modelling can begin.

Phase	Activity
2. Environment Integration	Establish secure connections to the client's sensor and SCADA data feeds, or agree an interim data-sharing arrangement, and provision the infrastructure required to run the system within the client's environment.
3. Model Calibration and Pilot	Calibrate the classification and Remaining Useful Life models against the client's specific equipment and operating conditions, and run a time-boxed pilot at a single site to validate performance before wider rollout.
4. Staff Training and Change Management	Train the client's maintenance planners and technicians on the dashboard, alert workflow, and revised scheduling process, and incorporate their feedback into the interface.
5. Go-Live and Ongoing Support	Transition the pilot site to a monitored production service with defined uptime expectations, then extend deployment to additional sites under a support and maintenance agreement.

KPC is the intended first deployment under this methodology. Phases 1 through 3 will be scoped and demonstrated using KPC as the anchor client within the fellowship's Weeks 7 through 12 delivery period, culminating in the Capstone Showcase.

7. The KPC Ask

To progress from a validated synthetic-data prototype to a system KPC can adopt with confidence, the team requests the following:

- Assignment of a technical contact within KPC's engineering team, to validate the sensor and failure-signature assumptions used in the synthetic dataset.
- Access to aggregated historical pump failure records, such as failure counts per station per year, to calibrate the synthetic dataset against actual operating conditions.
- A structured consultation with a maintenance planner, to confirm the proposed dashboard workflow aligns with existing scheduling practice.
- Information on KPC's existing business intelligence and dashboard infrastructure, to ensure compatibility with current systems.
- An introduction to the team responsible for pump maintenance budgeting, to establish sponsorship for the Phase 1 pilot.